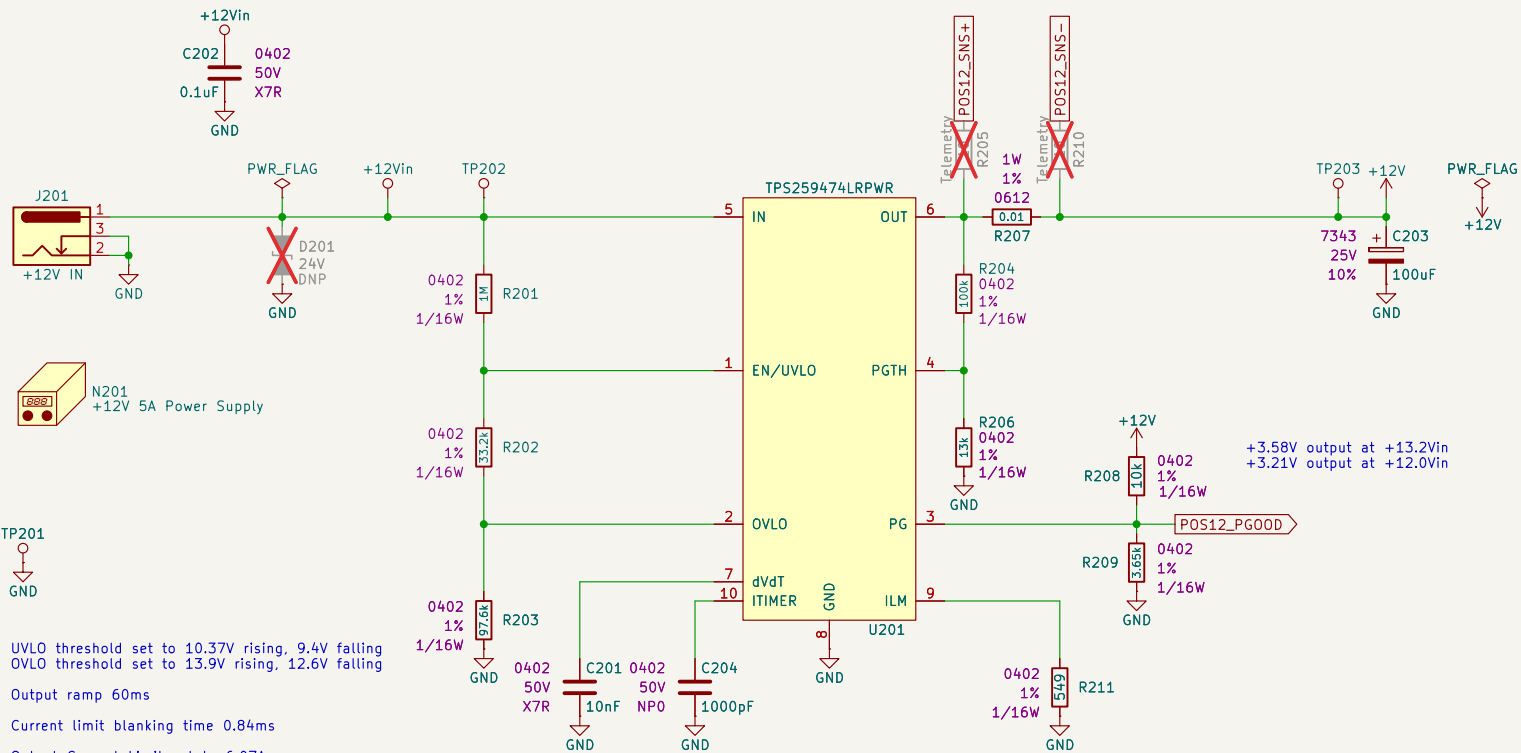


	1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch	Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Buffers File: led_panel_buffers.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch LED Panel Connectors File: led_panel_connectors.kicad_sch				
B						
C						
D						
	1	2	3	4	5	



Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A @ 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

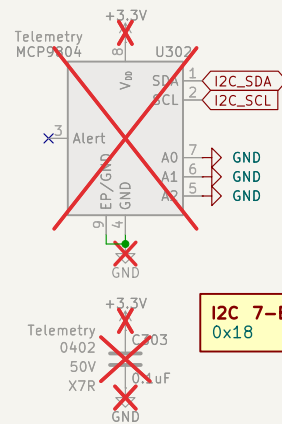
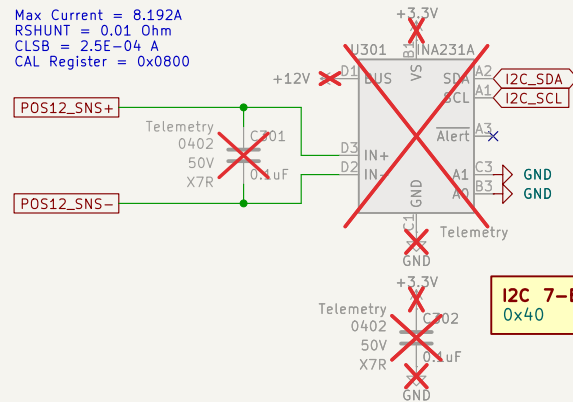
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 File: POS12_power_input.kicad_sch

Title:

Size: A Date: 2026-10-08
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Max Current = 8.192A
RSHUNT = 0.01 Ohm
CLSB = 2.5E-04 A
CAL Register = 0x0800

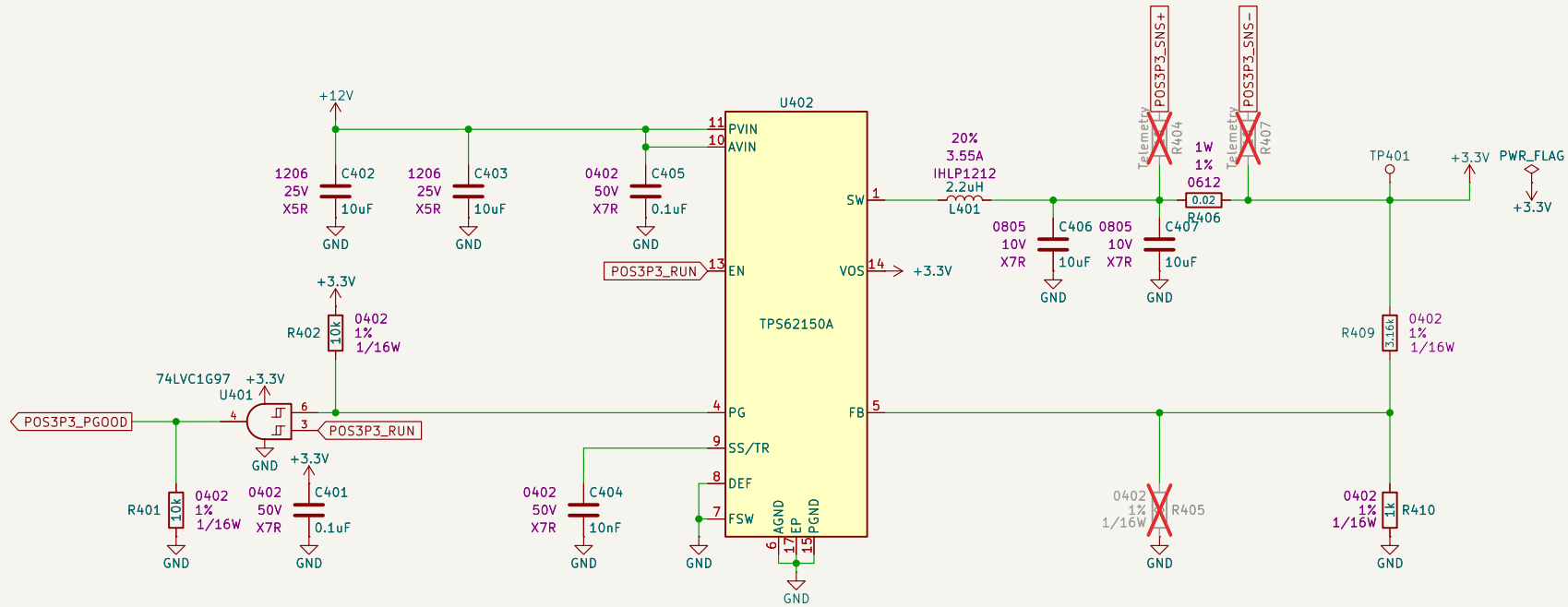


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File: POS12_telemetry.kicad_sch

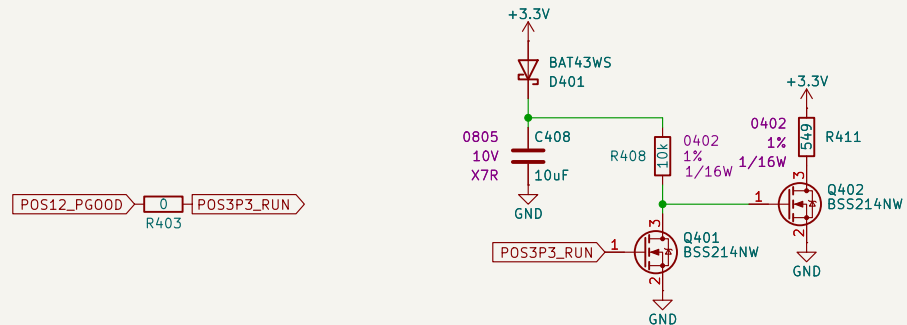
Title:

Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

Rev: A
Id: 3/21



Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	6	6.6mA
MCP6024	5.4mA	3	16.2mA
Analog_Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V_Reference	0.2mA	1	0.2mA
PCA9536_GPIO_Expander	0.29mA	1	0.29mA
I2C_Buffers	8.3mA	3	24.9mA
Octal_Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary_Encoder_Module	50mA	1	50mA
LEDs	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap_Touch_Sensor	0.729mA	1	0.729mA
Total		0.579W	175.48mA



Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

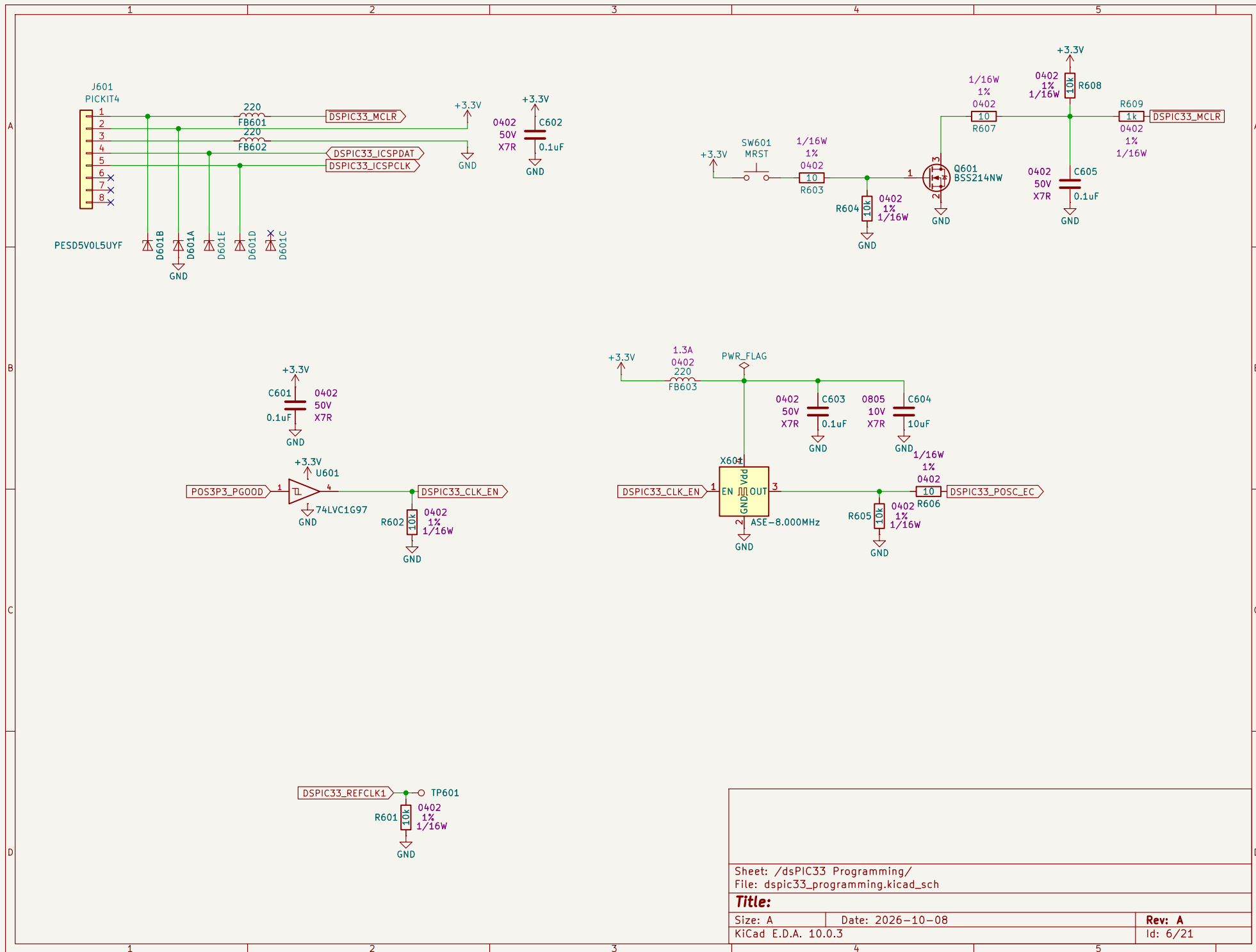
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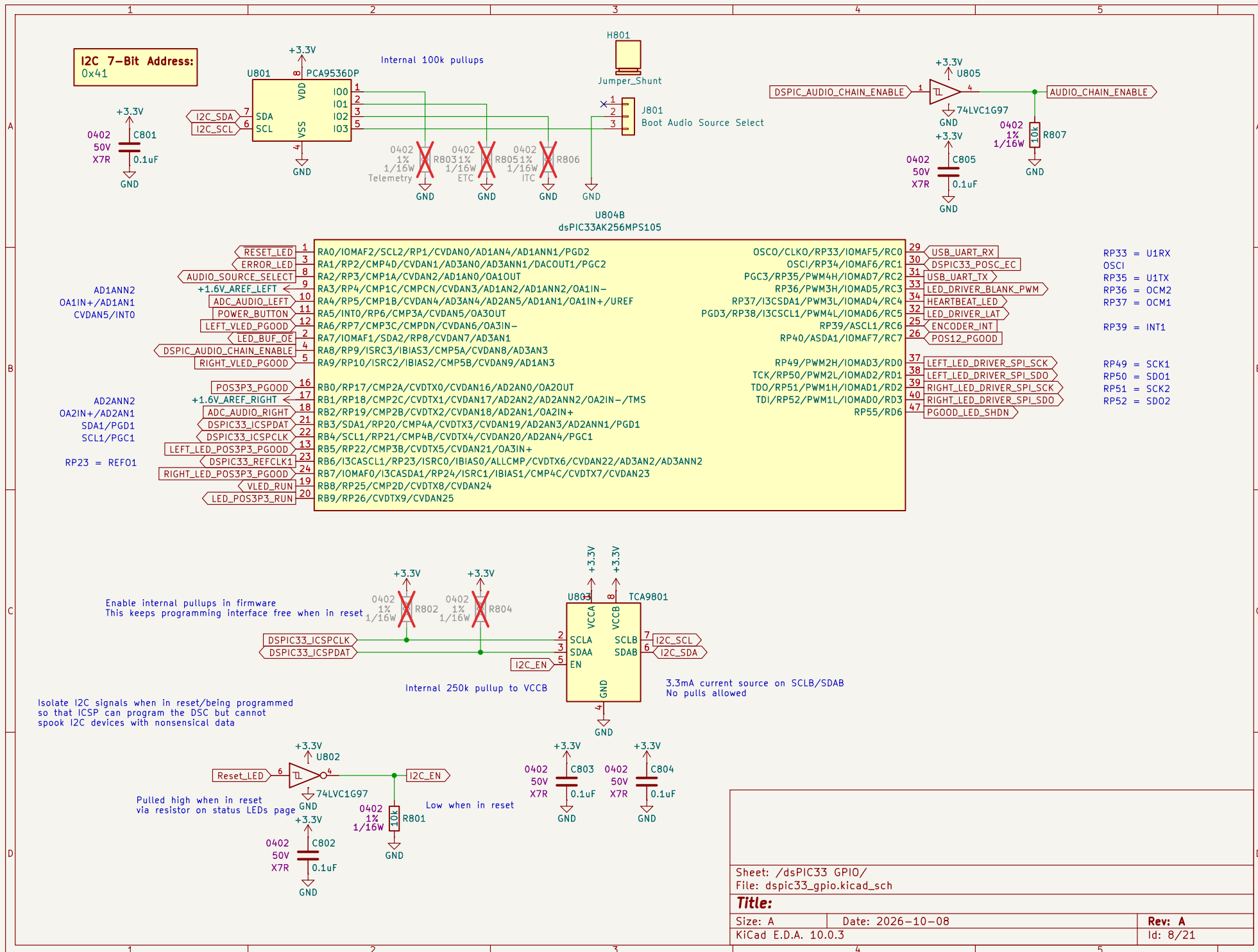


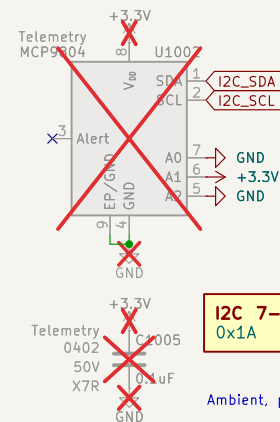
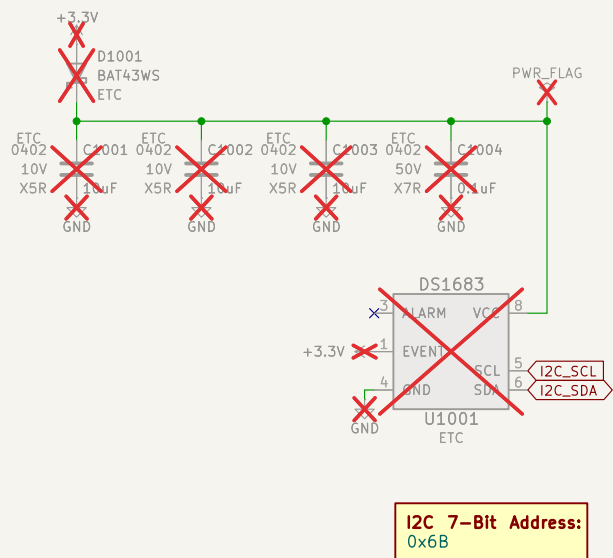
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Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

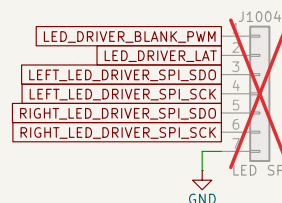
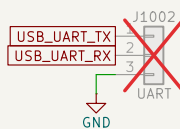
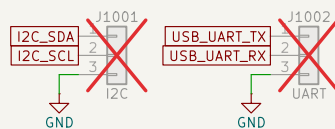
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Id: 6/21





Ambient, place away from power components

Test Points

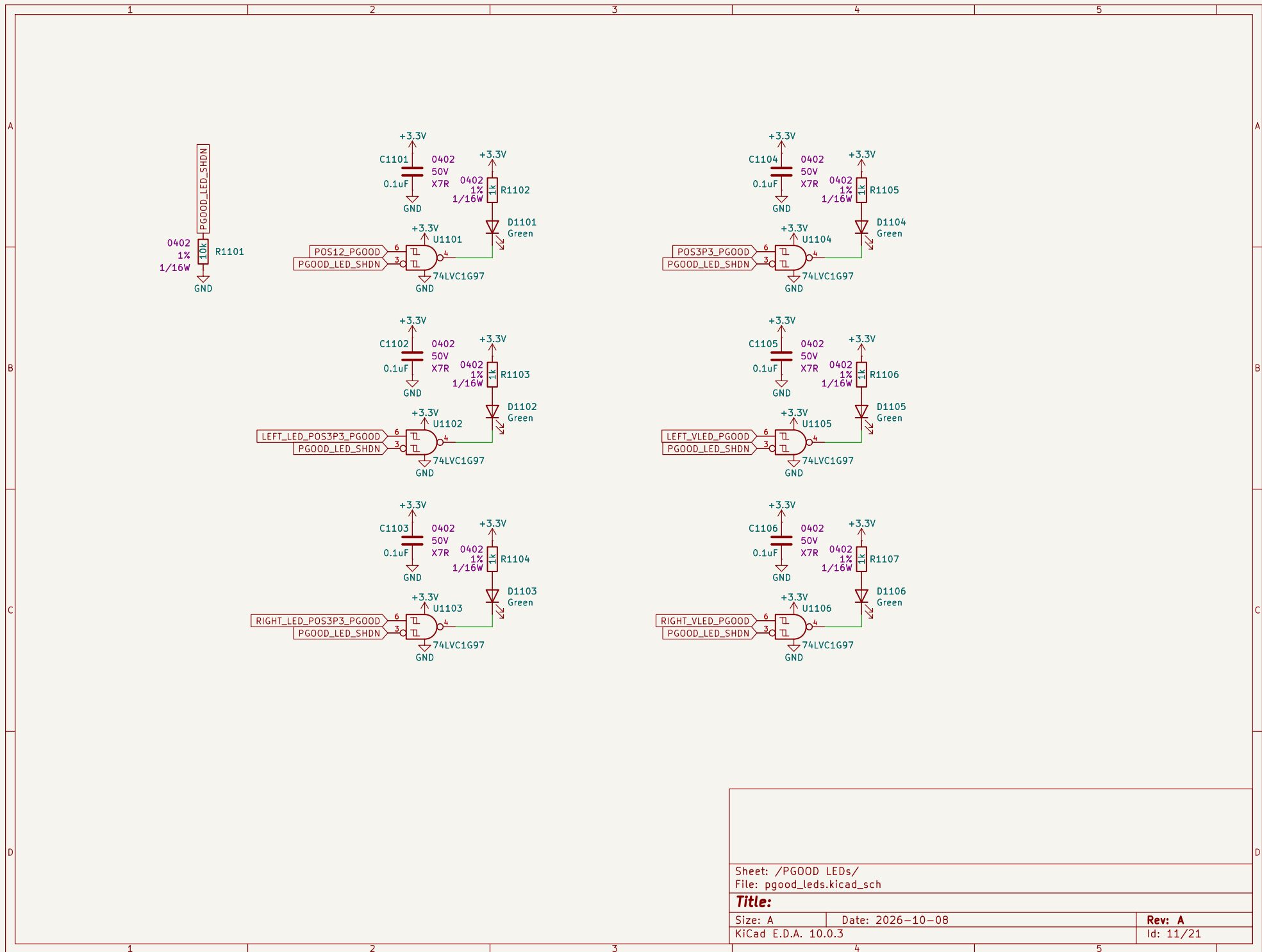


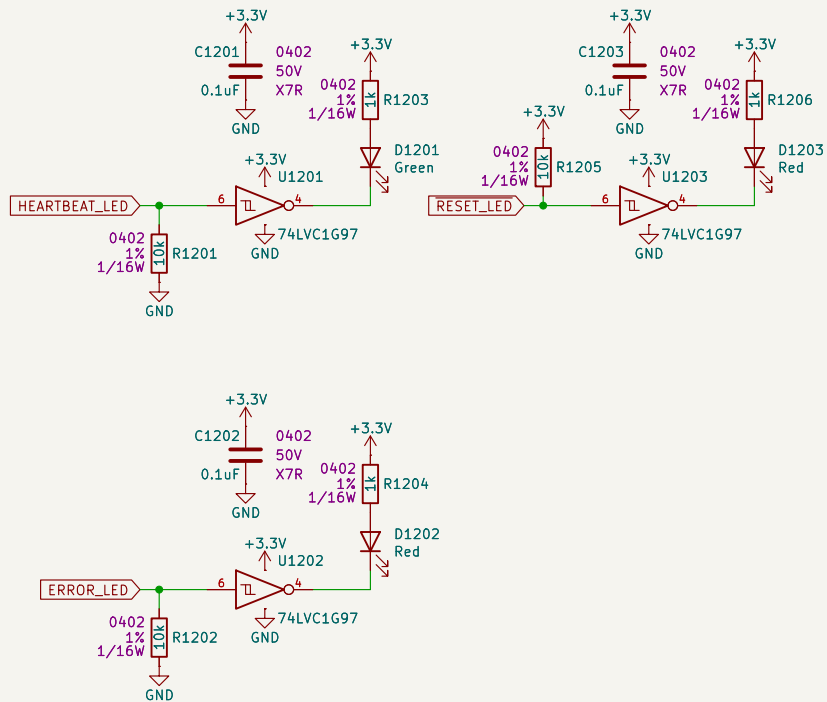
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File: platform_etc.kicad_sch

Title:

Size: A Date: 2026-10-08
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Rev: A
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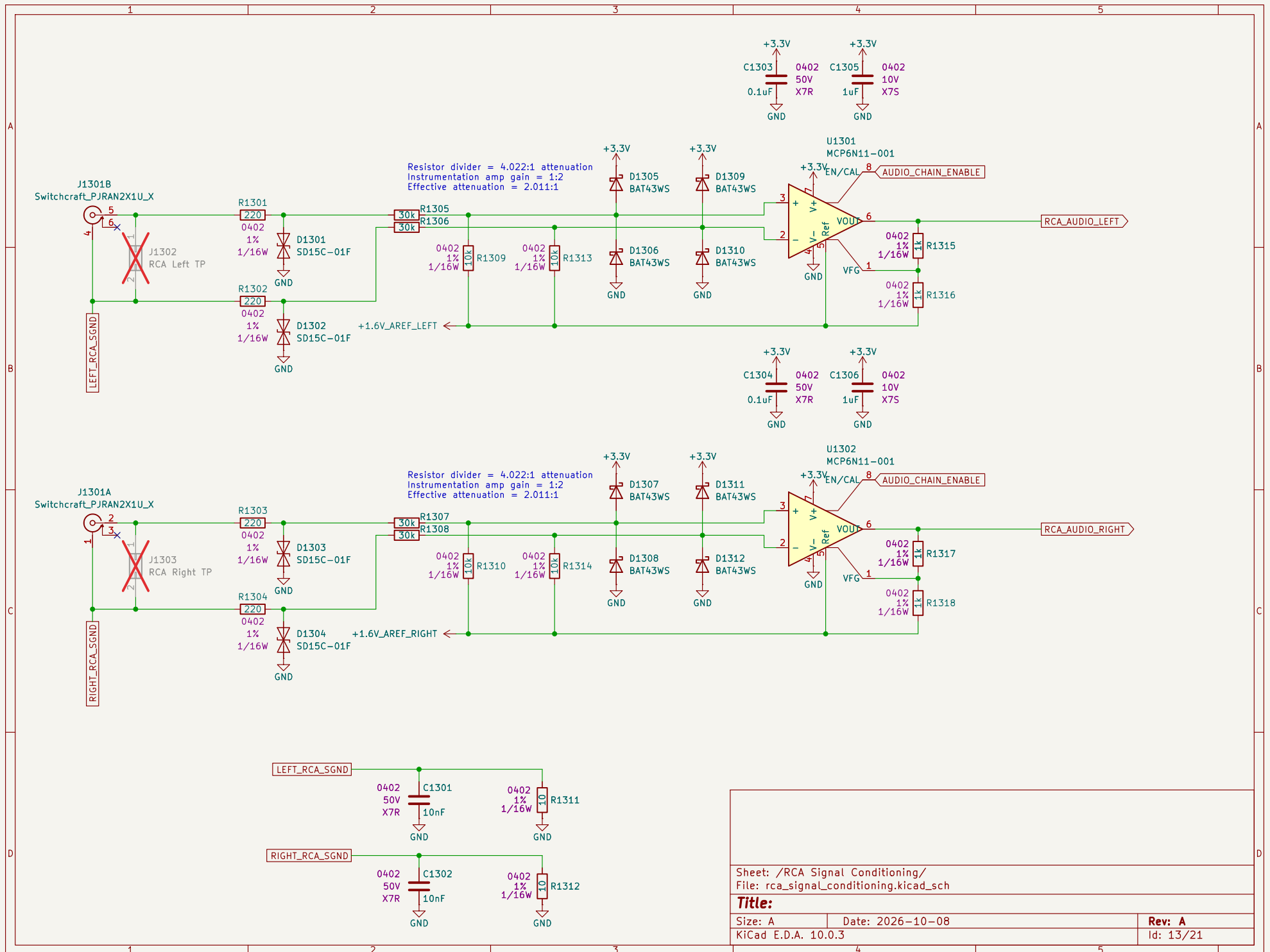


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

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Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch

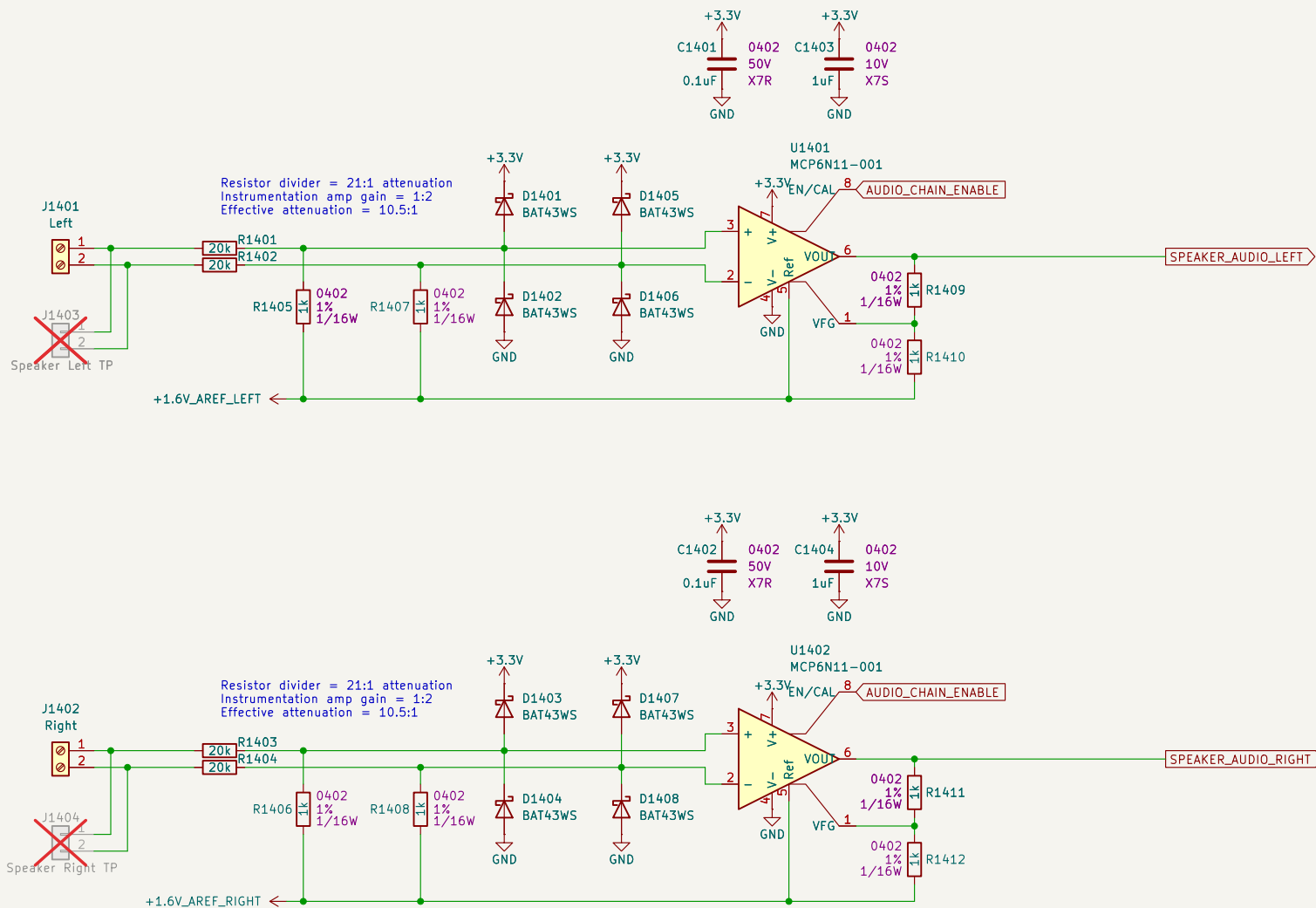
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KiCad E.D.A. 10.0.3

Rev: A

Id: 13/21



Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

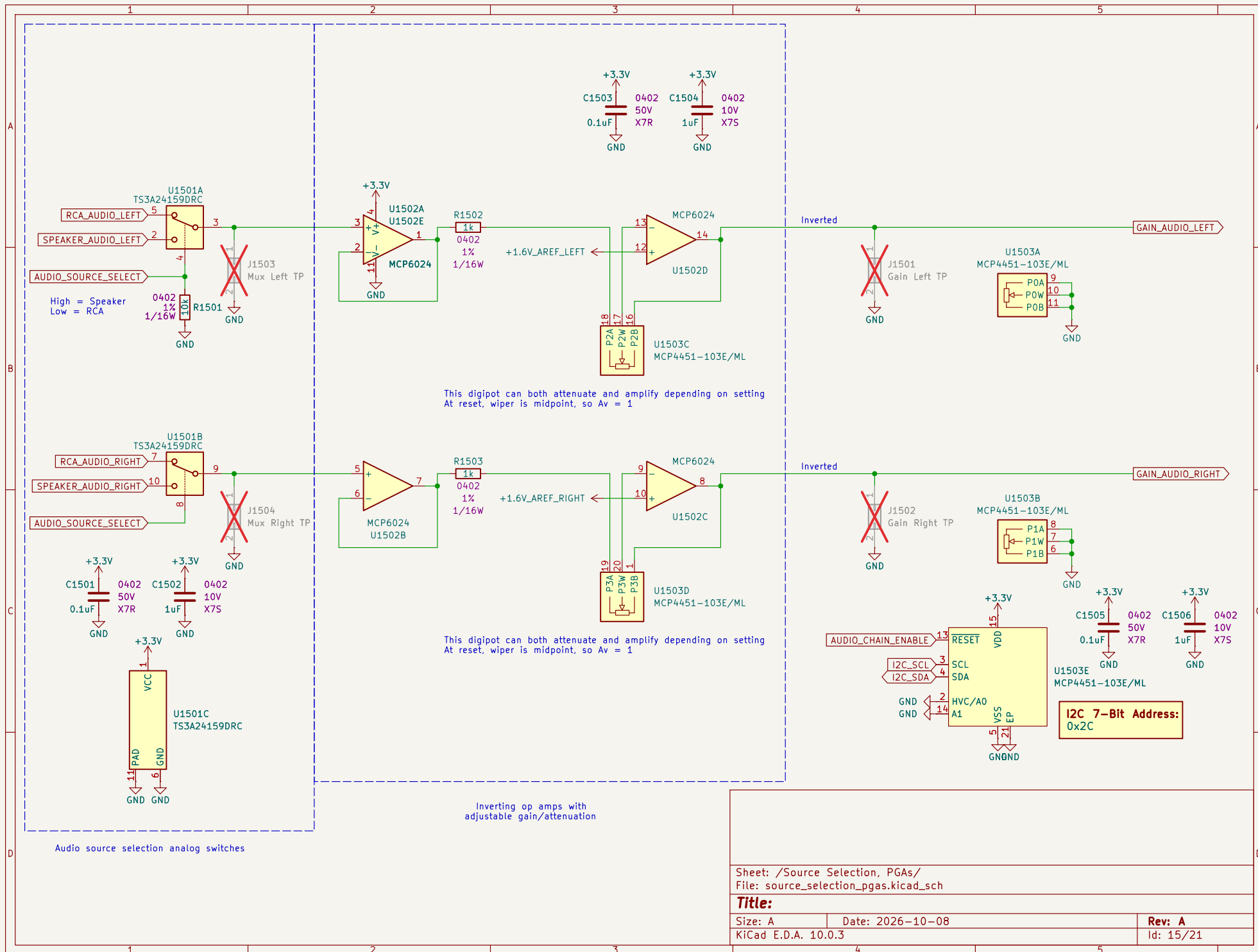
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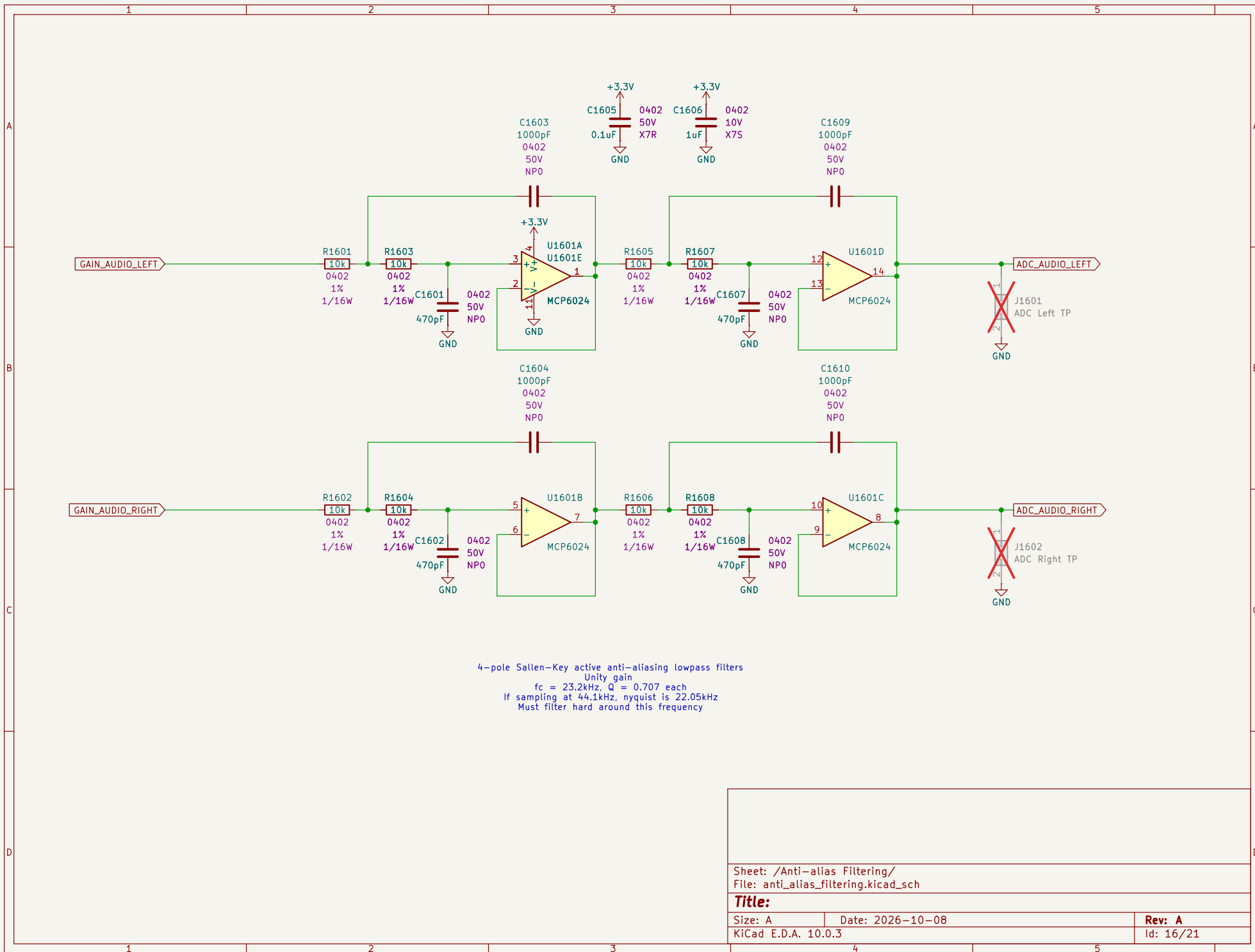


Sheet: /Source Selection, PGAs/
 File: source_selection_pgas.kicad_sch

Title:

Size: A Date: 2026-10-08
 KiCad E.D.A. 10.0.3

Rev: A
 Id: 15/21

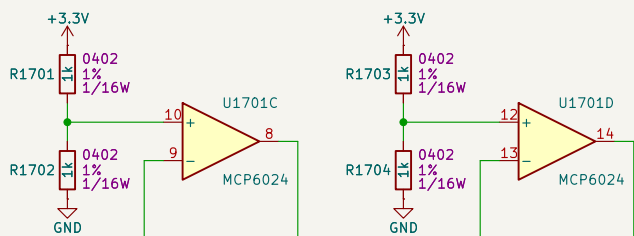
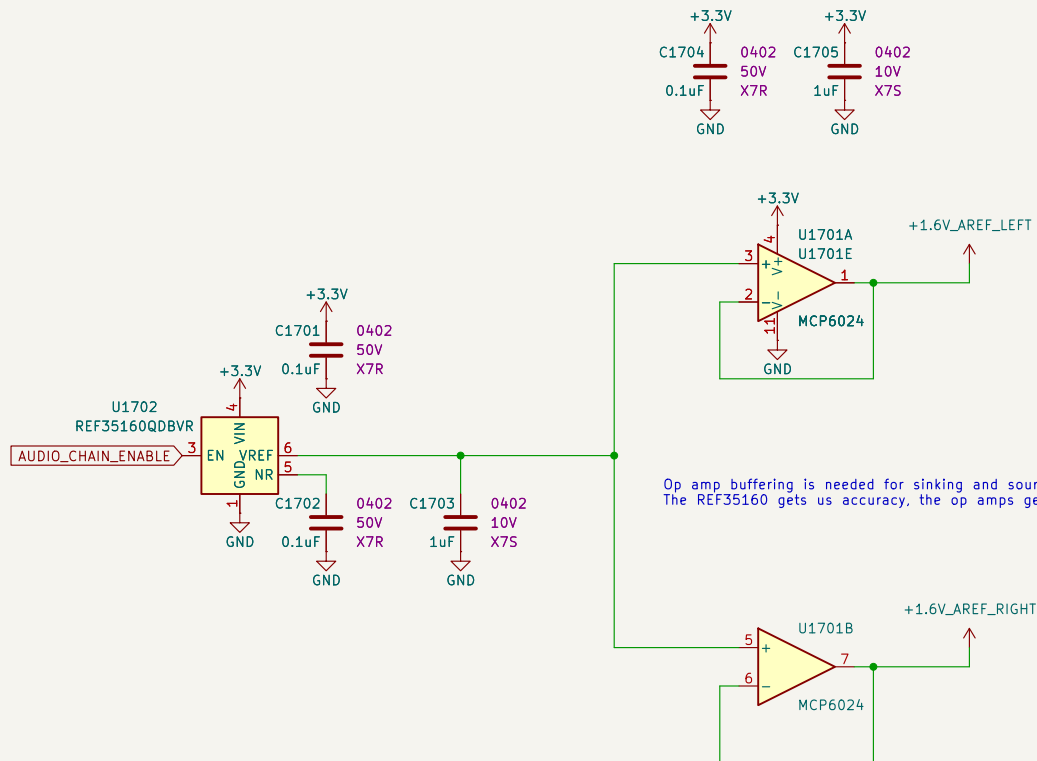


Sheet: /Anti-alias Filtering/
File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

Rev: A
Id: 16/21

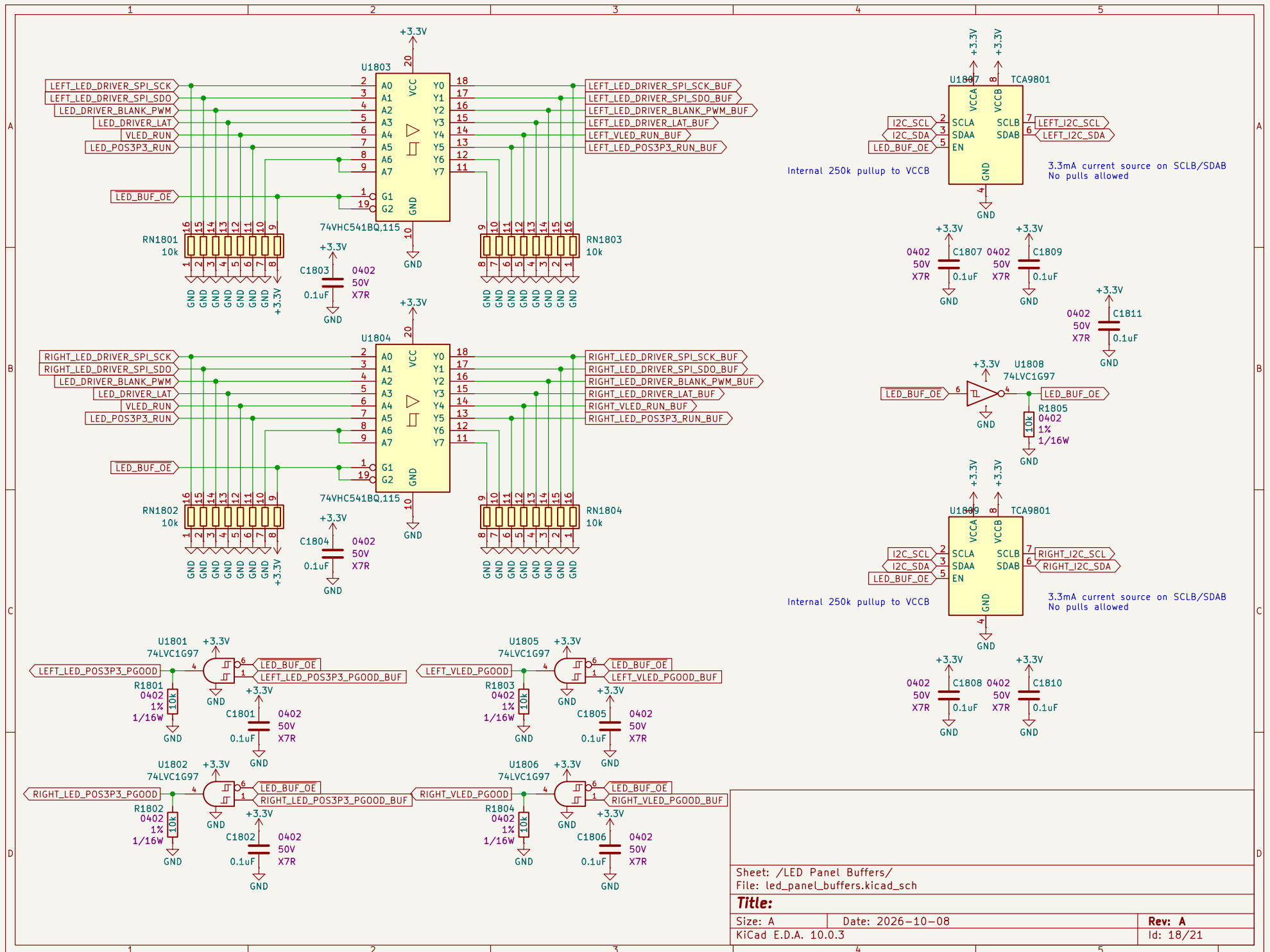


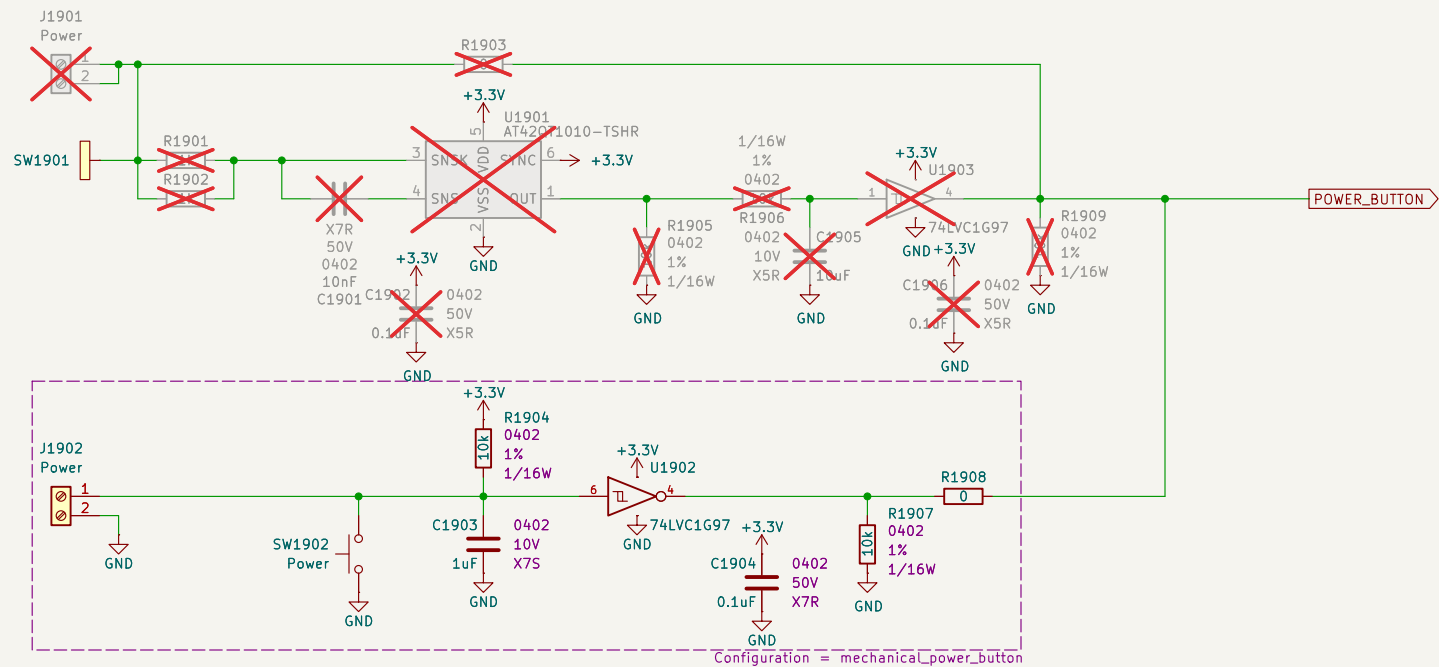
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File: audio_voltage_references.kicad_sch

Title:

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KiCad E.D.A. 10.0.3

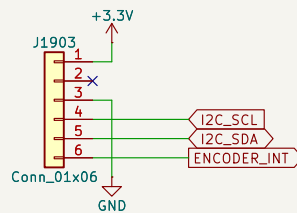
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N1901
Adafruit Rotary Encoder Module

I2C 7-Bit Address:
0x36



Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

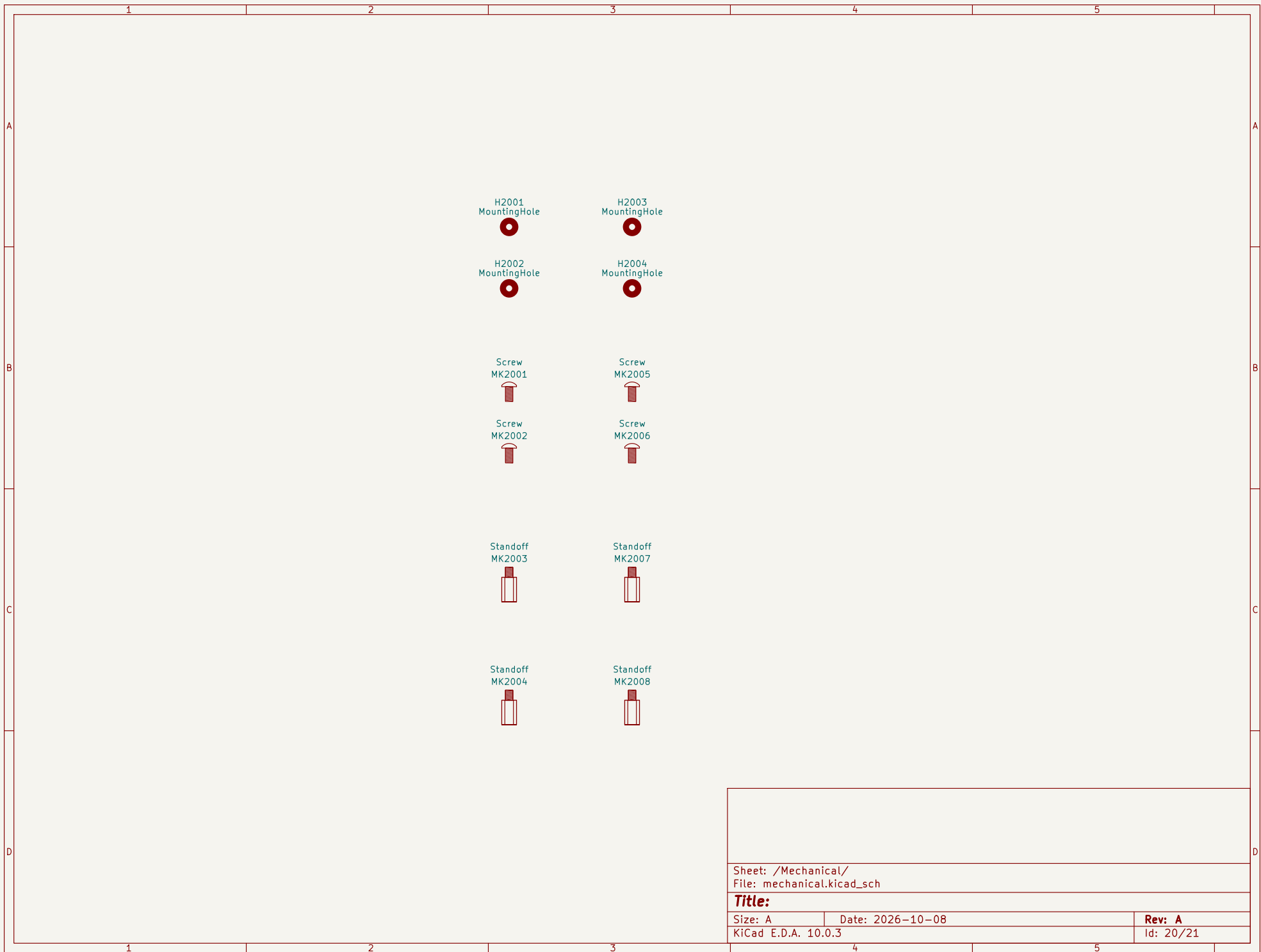
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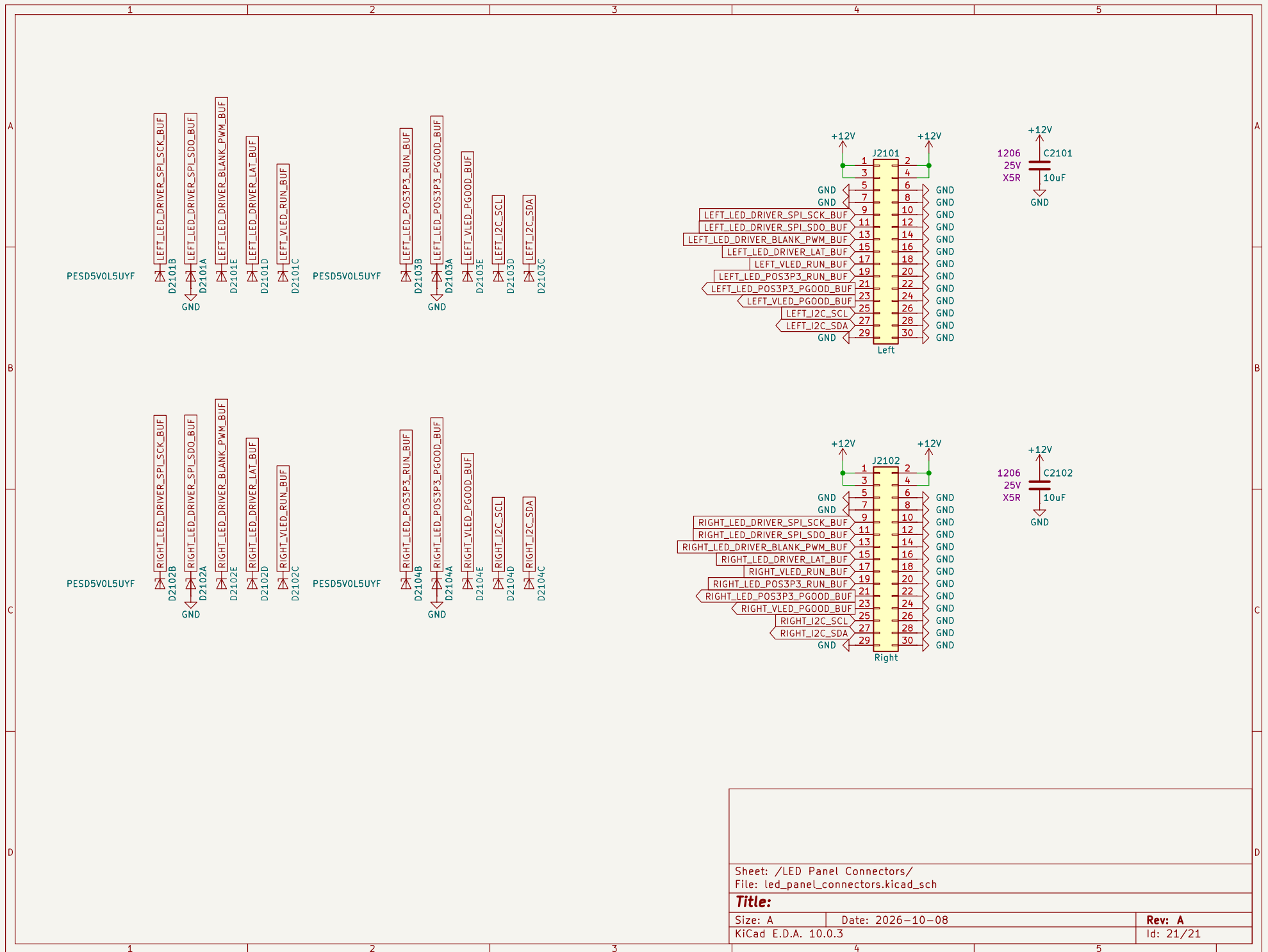
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KiCad E.D.A. 10.0.3

Rev: A

Id: 19/21





Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title:

Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

Rev: A
Id: 21/21